PADHAKU
Curiosity · Class 6 Science



CHAPTER 10

Living Creatures: Exploring their Characteristics

NCERT Exercise Solutions

Activity 10.1: Let us record	1 Q
10.1 What Sets the Living Apart from the Non-living?	14 Q
10.2 Essential Conditions for Germination of a Seed	8 Q
10.3 Growth and Movement in Plants	2 Q
10.4 Life Cycle of a Plant	2 Q
10.5 Life Cycle of Animals	13 Q
Let us enhance our learning	9 Q
Learning further	5 Q
Let us create	1 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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Activity 10.1: Let us record

Activity 10.1 List things around you (pencil, book, pigeon, car, plant, any other) in Table 10.1.

Write your guess (living/non-living) in column II and a reason in column III. Later, fill columns IV (correct answer) and V (reason for the correct answer).

💡 SAMPLE ANSWER

Name	My guess	Correct answer	Reason
Pencil	Non-living	Non-living	Does not grow, eat, breathe or reproduce
Book	Non-living	Non-living	Does not grow, breathe or respond
Pigeon	Living	Living	Moves, eats, grows, breathes, lays eggs
Car	Non-living	Non-living	Moves only when driven; does not grow, breathe or reproduce
Plant	Living	Living	Grows, needs food, respires, responds, makes seeds
Earthworm (any other)	Living	Living	Moves, eats, grows, breathes, reproduces

Conclusion: Living beings show all the features of life; non-living things do not.

⭐ 10.1 What Sets the Living Apart from the Non-living?

? Why do you think a pencil is non-living but a pigeon is living? What are the differences between living beings and non-living things? What similarities do the living beings share?

✔ ANSWER

A pigeon eats, grows, breathes, moves on its own, responds and lays eggs. A pencil does none of these, so it is **non-living**.

Similarities of living beings: they move, need food, grow, respire, excrete, respond to stimuli, reproduce and finally die.

? A car moves on the road. Does it mean that a car is living? List the tasks that you can do but a car cannot.

💡 SAMPLE ANSWER

No. A car moves only when someone drives it using fuel; it cannot move on its own.

Tasks I can do but a car cannot:

- eat food and grow
- breathe
- remove waste (sweat, urine)
- feel pain or heat and react

- produce young ones

? **A car does not grow. Does it mean it is non-living? Which characteristics have you used to classify a car as non-living?**

✔ ANSWER

Not growing alone is not enough; we check all the features together.

A car does not grow, does not need food, does not breathe, does not excrete, does not respond to stimuli and does not reproduce. So it is **non-living**.

? **Can we consider movement as a characteristic to differentiate between the living and the non-living? List five things around you that can move on their own. Are all of them living just because they move? Plants do not move from place to place; do you consider them living?**

💡 SAMPLE ANSWER

Things that move on their own: a dog, a cat, an ant, a bird, a fish.

Movement alone is not enough: a car, a fan or a clock also move, but they are non-living. We must check all features of life.

Yes, plants are living. They grow, respire, make seeds, and show some movements, e.g. flowers open, Drosera leaves trap insects, climbers wind around a support.

? **Can you wear the dress you wore four years ago? Can we consider growth as a characteristic of living beings?**

✔ ANSWER

No, because my body has grown bigger.

Yes. All living beings (humans, animals, plants) grow. This is a characteristic of living beings.

? **List five living beings that require food to grow.**

💡 SAMPLE ANSWER

Human, cow, dog, sparrow, mango tree (plants make their own food).

? **Count the number of breaths you take per minute after a normal walk, after a run, and after a few dance steps. Do you notice any difference? Do you notice breathing in animals like dogs, cats, cows and buffaloes?**

💡 SAMPLE ANSWER

Situation	Breaths per minute (sample)

After a normal walk	18
After a few dance steps	25
After a run	32

Yes, we breathe **faster** after running or dancing than after walking, because the body needs more air during hard work.

Yes, when animals rest, their abdomen moves up and down as they breathe.

? Do plants also respire?

✔ ANSWER

Yes. Leaves have tiny pores called **stomata** through which air goes in and out. All living beings respire.

? List three stimuli and your body's instant response to them.

💡 SAMPLE ANSWER

Stimulus	Response
Stepping on a thorn	Pull the foot back quickly
Touching a hot cup of tea	Take the hand away at once
Bright light in the eyes	Close the eyes / blink

? Find a few more plants in your neighbourhood which fold their leaves after

sunset.

💡 SAMPLE ANSWER

Tamarind (imli), gulmohar, babool (acacia), shami and pea plants fold their leaves after sunset.

? Why do the leaves of chhui-mui and amla plants respond in this way? Which stimulus could be responsible for their behaviour?

✔ ANSWER

Plants are living, so they **respond to stimuli**.

- Chhui-mui: the stimulus is **touch**.
- Amla: the stimulus is the **absence of light (darkness)** after sunset.

? List young ones of five different animals. Have you seen young ones of non-living things such as a pencil, a chair or an electric bulb?

✔ ANSWER

Animal	Young one
Cat	Kitten
Dog	Puppy
Cow	Calf
Hen	Chick
Frog	Tadpole

No. Non-living things do not produce young ones. Only living beings **reproduce**.

? Why is reproduction necessary?

✔ ANSWER

Reproduction is necessary for the **continuity of life**: it produces new ones of the same kind, so the kind does not die out.

? In which category would you place a seed—living or non-living? Why?

✔ ANSWER

A seed is **living**. It is alive but resting. When it gets water and air, it germinates, grows into a plant and later makes new seeds.

⭐ 10.2 Essential Conditions for Germination of a Seed

? What conditions do you think are required for seed germination? How will you investigate whether these conditions have an effect on germination?

💡 SAMPLE ANSWER

Water, air and maybe sunlight.

We can sow seeds in pots, give each pot a different condition (no water, too much water, darkness, all conditions) and see which seeds sprout.

Activity 10.2 **Four pots with bean seeds, kept for 15 days:**

- **Pot A: no water, in direct sunlight**
- **Pot B: excess water (water always above the soil), in direct sunlight**
- **Pot C: slightly moist soil, in a dark place**
- **Pot D: slightly moist soil, in direct sunlight**

Fill Table 10.2: availability of air, sunlight and water; prediction; observation; reason.

✔ ANSWER

Pot	Air	Sunlight	Water	Germination	Reason
A	Yes	Yes	No	No	No water to soften the seed coat

B	No	Yes	Yes (excess)	No	Water fills soil spaces, so no air reaches the seeds
C	Yes	No	Yes	Yes	Air and right amount of water are present
D	Yes	Yes	Yes	Yes	Air and right amount of water are present

Seeds germinate in **pots C and D**.

? **Is sunlight necessary for germination? Do the seeds in all pots receive air, water and sunlight? Is there a pot where air is not available, and why? What happens to the seeds in the pot with excess water? Which seeds receive both air and water? In which pots do seeds germinate?**

✔ ANSWER

- **No**, sunlight is not necessary: seeds in the dark pot C also germinate.
- No, all pots do not get all three. Pot A gets no water; pot C gets no sunlight.
- In **pot B**, air is not available because the excess water fills the spaces between soil particles.
- Seeds in pot B do not germinate (they may rot).
- Seeds in **pots C and D** get both air and water.
- Germination is seen in **pots C and D**.

? **Do your observations match your predictions? Based on your observations, state the conditions which favour seed germination.**

💡 SAMPLE ANSWER

Mostly yes (I had guessed that sunlight is needed, but seeds in the dark pot also germinated).

Conditions that favour germination: the **right amount of water** and **air**.

? **Which of the following are essential for seed germination—air, water and sunlight? Why do seeds require these conditions? Will the absence of one or more of these conditions affect germination?**

✔ ANSWER

Air and water are essential; sunlight is not needed for bean seeds.

- Water softens the seed coat and helps the tiny embryo inside develop into a plant.
- Seeds use the air in the spaces between soil particles.

Yes. If water or air is missing, the seeds will not germinate.

? **What other conditions do you think would affect seed germination?**

💡 **SAMPLE ANSWER**

- Suitable temperature (not too hot or too cold)
- Healthy seeds (not damaged or eaten by insects)
- Light or darkness for some seeds (Coleus and Petunia need light; Calendula and Zinnia need darkness)
- Depth of sowing and loose soil

? **How would you now categorise a seed, as living or non-living?**

✔ **ANSWER**

As **living**. Given water and air, it germinates and grows into a new plant.

? **In Activity 10.1, what characteristics made you place plants in living beings? Do plants**

show growth in Activity 10.2? Are there any other characteristics of living beings that these plants show?

✔ ANSWER

Plants grow, need food, respire, excrete, respond to stimuli and reproduce.

Yes, the seedlings in Activity 10.2 grow bigger day by day.

They also show movement (the shoot grows up, roots grow down) and they use air (respire).

⭐ 10.3 Growth and Movement in Plants

? **How do plants respond to sunlight? Does sunlight affect the direction of growth? In which direction would the root and shoot grow if the plant is placed inverted? How would you design an activity to find out?**

✔ ANSWER

The shoot grows and bends **towards sunlight**.

If the plant is placed upside down, the root bends to grow **downwards** and the shoot bends to grow **upwards**.

We can fix seedlings upright, inverted and with light from one side only, and watch them (Activity 10.3).

Activity 10.3 **Seedlings fixed on glass plates in beakers: A upright, light from all**

directions; B inverted, light from all directions; C upright, light from one direction only (box with a hole).

Fill Table 10.3. What is the direction of growth of root and shoot in A, B and C? Do your predictions match? What do you conclude?

✔ ANSWER

Beaker	Plant	Shoot grows	Root grows
A (light from all sides)	Upright	Upwards	Downwards
B (light from all sides)	Inverted	Bends and grows upwards	Bends and grows downwards
C (light from one side)	Upright	Towards the light (the hole)	Downwards

Conclusion: Shoots grow upwards and move towards sunlight; roots always grow downwards.

⭐ 10.4 Life Cycle of a Plant

Activity 10.4 Plant a bean seed, observe it for three months and record changes in Table 10.4.

How long does it take for any change to occur? After how many days does

the first flower appear? After some parts of the flower dry, is there further growth? Which structure do the remaining parts of the flower develop into? Can you notice a pod with seeds? What happens to the plant after fruits with seeds are formed?

💡 **SAMPLE ANSWER**

Day (approx.)	Observation
Day 1	Seeds are sown
Day 4–7	Seed germinates; root and shoot appear
Week 2–3	Leaves appear; plant grows taller
Week 5–7	First flowers appear
Week 7–9	Petals dry; pods (fruits) with seeds form
Week 10–12	Plant turns yellow and dries

- The first change (germination) appears in a few days.
- After some parts of the flower dry, the remaining part grows into a **pod (fruit)** with **seeds**.
- After the pods form, the plant slowly becomes yellow, dries and dies.

? What changes do you observe after the fruits are formed? Does the plant become yellow and dry even when you continue watering it?

✔ ANSWER

After the fruits form, the leaves turn yellow, the plant stops growing and dries up.

Yes, it dries even with watering, because its life cycle is complete. The seeds from its pods can grow into new bean plants.

⭐ 10.5 Life Cycle of Animals

? Have you ever observed how animals grow over time? Draw sketches of their young ones and name them.

💡 SAMPLE ANSWER

(Draw them.) Examples: cow – calf, hen – chick, dog – puppy, frog – tadpole, butterfly – caterpillar.

? We are advised not to allow water to stagnate anywhere. Why? Does stagnant water have any relation with mosquitoes laying eggs?

✔ ANSWER

Yes. Female mosquitoes lay eggs on or near stagnant water. The larvae and pupae grow in that water.

So stagnant water helps mosquitoes breed, and mosquitoes spread malaria, dengue and chikungunya.

? Discuss the measures one can take to prevent breeding of mosquitoes. What differences do you observe in the shape of larvae and pupae?

💡 SAMPLE ANSWER

Measures:

- Do not let water collect in coolers, pots, tyres or open containers; empty them every week.
- Cover water tanks and containers.
- Spray kerosene oil on water that cannot be removed.

Shapes: The larva is long and thin like a worm. The pupa is short, with a big rounded head part, like a comma.

? Mosquito larvae and pupae repeatedly come to the water surface. What can be the reason?

✔ ANSWER

They live in water but need **air to respire**, so they come to the surface to take in air.

? How can the life cycle of a mosquito be disrupted? Why does mother spray kerosene

oil on stagnant water?

✔ ANSWER

By not letting water stagnate, so mosquitoes cannot lay eggs.

Kerosene oil forms a thin layer on the water. It separates the water from the air, so larvae and pupae cannot take in air and **they die**.

Activity 10.5

How will you decide which stage (larva or pupa) comes immediately after the egg?

Without separating them from the container of puddle water, how would you design an activity to decide which stage gives way to the next?

💡 SAMPLE ANSWER

Plan without separating them:

- Step 1: Count the larvae and the pupae in the container on day 1.
- Step 2: Cover the container with a net so that adult mosquitoes cannot fly away.
- Step 3: Count larvae and pupae every day.
- Step 4: If the number of larvae goes down while the number of pupae goes up, larvae are changing into pupae.
- Step 5: Mosquitoes will come out from the pupae first.

Result: egg → **larva** → pupa → adult. The larva comes right after the egg.

Activity 10.6 **Observe the stages of a frog in Fig. 10.7 (A, B, C, D, E, F). How will you decide their sequence? Record the changes in Table 10.5 (D: 'It is similar to C but it has two legs.').**

💡 SAMPLE ANSWER

We decide the order by how much the body has changed: eggs first, then tadpoles with tail, then legs, then a frog.

Stage	Change seen
A and F	Eggs in a jelly (spawn) and embryo – no legs or tail
C	Tadpole with a tail, no legs
D	Similar to C but has two (hind) legs
Froglet	Four legs, tail getting shorter
Adult frog	No tail, strong legs

Order: **A/F → C → D → froglet → adult frog**. B and E are the froglet and the adult frog (check the picture in your book).

? **How are these eggs of a frog different from the other eggs that you may have seen?**

✔ ANSWER

Frog eggs are small, soft and have no hard shell. They are laid in water in a cluster held together by a white jelly (**spawn**). Hen eggs are big, laid one at a time and have a hard shell.

? Which stage has the shortest duration?

✔ ANSWER

The **egg (spawn) stage**: it changes into an embryo in about 3–4 days (Fig. 10.8).

? Is there a change in the habitat during the various stages in the life cycle of a frog?

✔ ANSWER

Yes. Eggs and tadpoles live only in water. Froglets live in water but begin to spend some time on land. Adult frogs live both in water and on land.

? How do the special features support that stage?

✔ ANSWER

- Jelly around the eggs keeps them together and protects them.
- The tail helps the tadpole swim in water.
- Strong legs help the adult frog jump and land, and live on land as well as in water.

? **Do you think that birds also show significant changes in the various stages of their life cycle?**

✔ ANSWER

No. A chick that hatches from an egg looks like a small adult bird. It only grows bigger and gets feathers; there is no larva, pupa or tadpole-like stage.

? **How does the life cycle of animals differ from that of plants?**

💡 SAMPLE ANSWER

- Plants start from a seed; most animals start from an egg (or are born as young ones).
- Plants grow in one place and produce flowers, fruits and seeds; animals move about.
- Some animals (mosquito, frog) change body shape greatly and even habitat; plants do not have such stages.

Let us enhance our learning

1 **List the similarities and differences in life cycles of plants and animals.**

✔ ANSWER

Similarities:

- Both are born, grow, become adults, reproduce and finally die.
- Both need food, air and water and pass through several stages.
- Reproduction keeps their kind going.

Differences:

- Plant: starts from a seed; stages are seed, germination, leaves, flowers, fruits with seeds.
- Animal: starts from an egg or as a young one; some (mosquito, frog) change shape and habitat a lot.

2 **Find examples for these conditions. If an example is not possible, explain why.**

S. no.	Does it grow?	Does it respire?
1.	No	No
2.	No	Yes
3.	Yes	No
4.	Yes	Yes

💡 **SAMPLE ANSWER**

S. no.	Example	Remarks
1.	Stone, pencil, chair	Non-living things
2.	A dry stored seed	It is living and respires slowly but does not grow until it gets water

3.	Not possible	Every living being that grows also respires; growth needs energy from respiration
4.	Plants, animals, humans	Living beings

3 **You have learnt that different conditions are required for seed germination. How can we use this knowledge for proper storage of grains and pulses?**

✔ ANSWER

Seeds need water (moisture) and air to germinate.

So grains and pulses should be **dried well** and stored in **dry, airtight containers** in a cool place. Without moisture they will not sprout or rot.

4 **A tail is present in a tadpole but it disappears as it grows into a frog. What is the advantage of having a tail in the tadpole stage?**

✔ ANSWER

A tadpole lives only in water. The **tail helps it swim** to find food and escape from enemies.

5 **Charan says that a wooden log is non-living as it cannot move. Charu says it is living because it is made of wood obtained from**

trees. Give your arguments in favour of or against the two statements.

✔ ANSWER

Charan is right that the log is non-living, but his reason is not complete: not moving alone does not make something non-living (plants do not move from place to place, yet they are living). The log is non-living because it does not grow, respire, excrete, respond or reproduce.

Charu is wrong. The wood came from a living tree, but once cut, the log shows no features of life. It is now dead/non-living.

6 **What are the similarities and distinguishing features in the life cycles of a mosquito and a frog?**

✔ ANSWER

Similarities:

- Both lay eggs in or near water.
- Their young ones (larva, tadpole) live in water and look very different from the adults.
- Both show big changes in body shape during their life cycle.

Differences:

Mosquito	Frog
Egg, larva, pupa, adult	Egg (spawn), tadpole, froglet, adult
Has a pupa stage	No pupa stage

Adult flies in the air	Adult lives in water and on land
Is an insect	Is an amphibian

7 **A plant is provided with all the conditions suitable for its growth, but the pot is kept lying along the ground (Fig. 10.9). Draw what you expect to see in the shoot and the root after one week. Write down the reasons.**

✔ ANSWER

(Drawing: the stem bends and points up; the root bends and points down.)

- The **shoot bends and grows upwards**, towards sunlight.
- The **root bends and grows downwards**.

Reason: shoots grow upward and towards light; roots always grow downward.

8 **Tara and Vijay set up the experiment shown in Fig. 10.10. What do you think they want to find out? How will they know if they are correct?**

💡 SAMPLE ANSWER

They want to find out whether seeds need **both air and water** to germinate (e.g. one seed with no water, one with water and air, one fully under water).

They will watch the seeds for a few days. If only the seed that gets both air and water sprouts, their idea is correct. (check the picture in your book)

9 **Design an experiment to check if temperature has an effect on seed germination.**

💡 **SAMPLE ANSWER**

- Take three bowls with moist cotton and put 10 bean seeds in each.
- Keep bowl A in a refrigerator (cold), bowl B at room temperature, and bowl C in a very warm place (e.g. near a heater, about 45 °C).
- Keep the cotton moist in all bowls; everything else is the same.
- Count the seeds that germinate in each bowl for 5–7 days.

Expected result: most seeds germinate at room temperature; few or none in the cold or very hot bowl. So **suitable temperature is needed** for germination.

📘 Learning further

1 **Make a field trip to a local garden. Interact with a gardener to learn about various conditions and the time required for the growth of various plants.**

💡 **SAMPLE ANSWER**

Ask the gardener: How much water does each plant need? Sun or shade? Which season to sow? How many days to flower and fruit?

Example notes: marigold – sunny place, flowers in about 2 months; tulsi – grows in about 1 month, little water.

2 **Can we grow plants without germinating their seeds? Explore and cite some examples.**

✔ ANSWER

Yes, from other parts of the plant:

- Potato – from the 'eyes' (buds) of the tuber
- Sugarcane, rose, money plant – from stem cuttings
- Bryophyllum – from buds on the leaf edges
- Onion, garlic – from bulbs; ginger – from its underground stem

3 **Observe the life cycle of five plants grown at home, school or a nearby garden. Create a picture book of their stages with the name and duration of each stage.**

💡 SAMPLE ANSWER

Project. For each plant (e.g. bean, pea, mustard, tomato, marigold) draw or paste pictures of: seed → germination → leaves → flowers → fruits/seeds, and write the days for each stage.

4 **Try to observe some of the stages in the life cycle of a butterfly or a moth. Are these stages similar to the stages in the life cycle of a mosquito?**

✔ ANSWER

A butterfly goes through **egg → larva (caterpillar) → pupa → adult**.

Yes, these four stages are similar to those of a mosquito. But a butterfly lays eggs on leaves, not in water.

5 **Would the environment affect the life cycles of insects? Explore and list the factors that affect the life cycles of insects.**

💡 SAMPLE ANSWER

Yes. Factors:

- Temperature (warm weather speeds up the life cycle)
- Water and rainfall (mosquitoes need standing water)
- Humidity
- Availability of food
- Light / season
- Enemies (birds, frogs) and pollution or insecticides

🔭 Let us create

Poem **Add more lines to the incomplete poem. Include the stages in the development of a frog.**

In shaded and grassy bogs, / There lived a group of frogs. / ... One day sitting beside a reed, / Female frogs think it's time to breed .....

💡 **SAMPLE ANSWER**

In the pond they laid their eggs with care,
A jelly ball of spawn floating there.
Out came tadpoles with long thin tails,
Swimming fast like tiny whales.
Two legs came first, then two more grew,
The tail grew short, a froglet new.
Now grown-up frogs hop on the land,
And sing again — a happy band!

🎉 **Chapter 10 done! Revise the answers once more tomorrow.**